
Build the Geometry In, Don't Hope It Emerges: A Reflection-Symmetry Audit of a Skeleton World Model

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 The human body has an approximate left-right structure. Reflecting a pose hori-
2 zontally and exchanging anatomically corresponding left and right landmarks pro-
3 duces another valid pose representation, so any signed left-versus-right motion
4 quantity should reverse sign under that operation. We use this exact geometric
5 fact to audit what a self-supervised skeleton world model represents. The model
6 is a Skeleton Joint-Embedding Predictive Architecture (S-JEPA) trained on pose
7 sequences from the Gait Abnormality in Video Dataset (GAVD); the audit target
8 is a coordinate-derived comparison of left- and right-side motion, not a diagno-
9 sis or a validated clinical measurement. The quality-controlled cohort holds 625
10 pose sequences from 93 source videos, split at the source-video level so that no
11 clip, mirror, or augmentation from a test video can reach the matching training
12 set. A fresh encoder is trained for each of fifty registered fold-and-seed combina-
13 tions. Across every registered checkpoint the trained representation does not pre-
14 dict the laterality contrast better than an untrained encoder of the same architecture
15 (learned minus untrained source-balanced $R^2 = -0.02$, 95% CI $[-0.04, 0.00]$), it
16 is not reflection-equivariant, and training moves the representation slightly away
17 from the approximate equivariance present at initialization (q rises by 0.03, 95%
18 CI $[0.02, 0.05]$). Reflection augmentation does not repair any of this. The only
19 route to correct sign-reversing behavior is to build it into the read-out analytically,
20 which guarantees exact output antisymmetry but adds no learned content, since
21 the same construction on a random encoder does at least as well. For a skeleton
22 world model, bilateral symmetry is something you install, not something you get
23 for free from prediction.

1 Evidence status

24 The registered laterality v2.1 run is complete. All five outer folds, five optimization seeds, and
25 both training variants finished; the report was computed on real derived-pose data rather than the
26 synthetic smoke profile; and the two hard integrity gates—every registered fold and seed present,
27 and exact output oddness for the constructed read-out—pass. The results in Section 9 are the study's
28 empirical findings and are reported without any post-hoc change to the frozen protocol. An earlier
29 transductive experiment, in which an encoder was evaluated on sequences that had also entered its
30 training, is superseded; its numbers are omitted. Two things remain open: submission and artifact
31 release stay blocked while the institutional ethics determination, the data-use review, and the derived-
32 pose release review are unresolved, and every claim keeps the narrow boundary of Section 11.

2 Introduction

A world model earns its name by predicting or reconstructing parts of its input without being handed the downstream label. For a model of an articulated body, physical geometry offers a direct way to ask what the model has captured. The human skeleton has a bilateral structure, and one clean instance of it is reflection: flip the body horizontally and swap each named left landmark with its right partner (Figure 1). This is a group action on the pose representation, and it acts predictably on any quantity that measures a left-versus-right difference. If a representation has internalized the geometry of the moving body, that structure should show up in how the representation transforms.

We turn that expectation into a measurement. We ask whether an S-JEPA representation, trained with no laterality label, carries usable information about a signed left-versus-right motion contrast, and whether that information behaves correctly when the input is mirrored. Three statements are easy to run together and worth keeping apart. A mathematical wrapper can force a read-out to change sign; this is a construction and proves only correct implementation. A trained encoder followed by a probe can show useful sign-reversing behavior; this needs empirical support. The encoder tokens themselves can satisfy a direct reflection-equivariance test; this is a stronger and separate empirical question. We measure all three.

The workshop’s Physical Geometry theme asks for geometry-aware world models and honest understanding of articulated, deformable scenes, and our work also speaks to the cross-cutting call for evaluation protocols and responsible deployment. Our contributions are:

- a reusable **geometry audit** for articulated-body world models, organized around a known group action, that separates analytic output oddness, probe-level sign behavior, and intrinsic token equivariance so a wrapper’s guarantee is never mistaken for a learned property;
- a registered, **leakage-controlled evaluation protocol**: source-level nested cross-validation, fold-local representation training, uncertainty that keeps optimization-seed variation apart from source-sampling variation, and thresholds frozen before any result was seen;
- a **fully powered null**—self-supervised training produces neither the laterality contrast nor reflection-equivariant tokens, and it degrades the equivariance present at initialization, with reflection augmentation leaving the verdict unchanged;
- the **constructive alternative**—an odd/even parity decomposition in the read-out guarantees exact antisymmetric outputs, though the guarantee comes from the wrapper and holds just as well on an untrained encoder.

The last point is the title. To make a skeleton model respect a physical symmetry of the body, the reliable move is to build the symmetry in. Waiting for a generic predictive objective to discover it, at this scale and on this target, did not work.

3 Concepts and terminology

A **pose sequence** is a run of video frames represented by estimated body-landmark coordinates rather than pixels. Each sequence carries 33 BlazePose landmarks with horizontal, vertical, and depth coordinates. A **source video** is the recording that one or more sequences were extracted from; because a single video can yield several sequences, splitting sequences independently would put closely related material on both sides of a divide. An **encoder** turns a sequence into learned features; a **read-out**, or probe, is a simpler model fitted on those features to predict the target. The encoder follows a fixed schedule with no early-stopping validation set, while the read-out uses inner cross-validation to choose its ridge penalty. **Cross-validation** rotates which data are held out: an **outer fold** measures held-out performance and an **inner fold** makes a training-only choice. **Stratification** keeps dataset-annotation counts roughly balanced; those annotations only balance the split and serve one named confounding control, and never act as training labels.

4 Dataset inventory and cohort construction

GAVD is the Gait Abnormality in Video Dataset [Ranjan et al., 2025, GAVD project, 2026]. Its official distribution supplies annotations and public video links rather than raw media, and this study

works on derived pose archives, redistributing no source video or identity-bearing frame. The frozen inventory contains 666 annotation files describing 103 source videos. A matching pose archive exists for 642 annotations; the other 24 have none. Pre-specified quality-control rules retain 625 pose sequences from 93 source videos and exclude 17, which can happen when the target cannot be computed, too few selected landmarks are valid, or too few complete four-frame patches remain. These rules were fixed before any model result was inspected, and inclusion never depends on the sign or size of the target. Every retained sequence carries provenance—source annotation, pose-archive fingerprint, extraction version, pose-model identity, and visibility threshold—and a SHA-256 content digest binds the cohort table and arrays together so later stages reject any drift from the frozen protocol.

5 Coordinate-derived laterality target

The target compares movement on the two sides using five anatomical pairs: shoulders, knees, ankles, heels, and foot-index landmarks. The hips define the pelvis-centered frame but stay out of the target, because pelvis centering makes their two velocity magnitudes algebraically identical. For each pair the calculation keeps only **paired-valid transitions**, where both landmarks are observed at both ends of the same frame transition. With $m_{L,k}$ and $m_{R,k}$ the median left and right speeds for pair k ,

$$c_k = \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + \epsilon}, \quad y = \text{mean}_k c_k,$$

where ϵ prevents division by zero. Every pair must supply at least eight shared valid transitions, and all five must be usable. Positive y means relatively more left-side motion and negative y the reverse, in an arithmetic rather than clinical sense. Across the cohort y ranges over roughly ± 0.2 with a standard deviation near 0.06, so the signal to recover is subtle. Short missing stretches may be interpolated for encoder input, but interpolated coordinates never enter the target, and invalid coordinates come with a validity mask rather than a numeric placeholder. Automated tests confirm that changing the values under invalid entries leaves y unchanged, that mirroring the coordinates and mask negates y , and that mirroring twice restores the input.

6 Training, validation, and testing design

Splitting unit. All sequences from one source video go into the same fold, and mirrors and augmentations inherit that assignment, so no clip, mirror, or augmentation from an outer-test source can reach the matching outer-training set. GAVD supplies no persistent person identifier and we infer none, so two videos could show the same unidentified person; the split therefore supports held-out-source-video evaluation, a weaker and more honest claim than held-out-person evaluation.

Five outer folds. The 93 eligible sources are split into five stratified outer folds. Roughly four fifths train the encoder and fit the final read-out while the rest form the untouched test set; sequence counts vary because videos differ in size (Table 1). This is not one fixed split—each source is tested exactly once, so combining out-of-fold predictions gives one held-out prediction per sequence.

Table 1: Registered outer folds. Splitting and balancing operate on source videos, not sequences.

Fold	Train sources	Test sources	Train sequences	Test sequences
0	74	19	436	189
1	74	19	443	182
2	74	19	553	72
3	75	18	548	77
4	75	18	520	105

Four inner folds. Each outer-training set is divided into four inner folds; one holds 18–19 validation sources while 55–57 fit a candidate read-out, and the role rotates until every outer-training source has validated once. Inner fitting partitions hold 266–450 sequences and validation partitions 80–197. These folds choose the ridge penalty and nothing else—they do not select a checkpoint,

Bilateral reflection is a group action: flip the horizontal axis, swap left/right landmarks

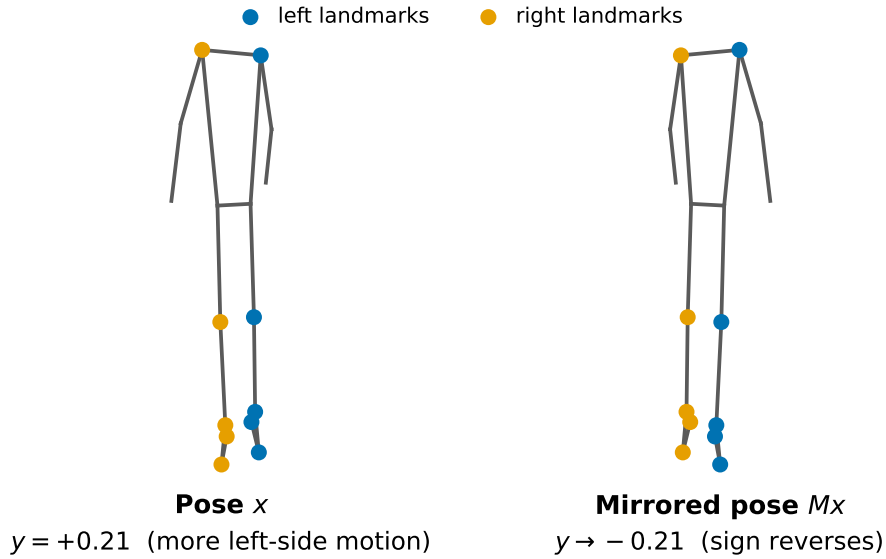


Figure 1: Bilateral reflection as a group action on a real GAVD-derived pose. Flipping the horizontal axis and swapping the anatomical left and right landmarks yields a valid pose representation, and the coordinate-derived target y reverses sign. The audit asks whether a learned representation respects this structure.

stop training, or change the schedule. After selection the read-out is refitted on all 74–75 outer-training sources and applied once to the 18–19 outer-test sources, with feature scaling, target scale, and neutral band all estimated from outer-training data only. If the outer-test targets helped pick the penalty, the final estimate would no longer describe untouched data.

7 Fold-local representation training

The representation model is an S-JEPA trained with a VICReg regularizer. It takes 64 time steps of 33 landmarks, groups time into four-frame patches, and uses a 96-dimensional embedding, a four-layer encoder, a two-layer predictor, and four attention heads. During training a fraction (0.6) of the sequence representation is masked and the model predicts the hidden latent content against an exponential-moving-average target encoder; neither the laterality target nor the dataset annotation enters this objective. Training is fold-local: for each fold, seed, and variant a new encoder starts from a recorded seed-matched initialization and sees only that fold’s outer-training sources, sampling a source uniformly before one of its sequences. The paper profile registers five seeds (42–46) and two variants—`vanilla` never reflects a training sample and `reflection_augmented` reflects one with probability 0.5. Within a seed the variants share initialization, source draws, masks, geometric views, optimizer schedule, and update count, with reflection on a separate random stream, so the pair stays matched. The design is $5 \times 5 \times 2 = 50$ fold-local encoders, each trained for 300 epochs with AdamW at learning rate 10^{-3} , weight decay 0.05, and batch size 20.

8 Held-out evaluation

For a given fold and seed, feature extraction uses the encoder that never saw that fold’s test sources. The primary lane fits a free linear ridge read-out to the learned representation. Comparison lanes include an identically structured untrained encoder (a paired initialization floor), a target-component self-consistency oracle (a verification, not a learned baseline), pooled and missingness features, constructed one- and two-pass antisymmetric read-outs, and direct encoder-token reflection tests. For

the parity constructions we split each feature into odd and even parts, $z^-(x) = (z(x) - z(Mx))/\sqrt{2}$ and $z^+(x) = (z(x) + z(Mx))/\sqrt{2}$, and give learned and paired-initial encoders equal-dimension single/free, odd/free, odd/zero-origin, and even/free lanes. The odd/zero-origin lane is the constructed repair; its parity and output oddness are imposed by design and never treated as emergence.

The principal measure is source-balanced R^2 , the fraction of target variation explained after each source video gets equal total weight. Mean absolute error, source-level sign accuracy, and mirror errors are secondary. Each checkpoint is scored on its own, with metrics and squared mirror errors formed within a checkpoint before the registered seeds are averaged, so signed errors cannot cancel. For the direct representation test, let $Z(x)$ and $Z(Mx)$ be target-encoder tokens and S the full 33-joint anatomical permutation acting on landmarks only; on tokens valid in both aligned views the strict error is

$$q = \frac{\|Z(Mx) - S Z(x)\|_C^2}{\|Z(Mx)\|_C^2 + \|S Z(x)\|_C^2},$$

with no channel fitting, rotation, sign search, centering, or read-out. Zero is exact equivariance, unrelated equal-energy representations sit near one, and a zero-energy collapse is rejected. Each learned checkpoint is paired with its own initial encoder, and the variants share that initialization for each fold and seed.

9 Results

The registered run supports none of the emergence claims and passes only the construction’s own integrity checks. We report the primary lane and its key comparisons; the pattern repeats for the reflection-augmented variant and for a high-coverage sensitivity subset (600 sequences from 91 sources at ≥ 0.9 authorized coverage), where the primary source-balanced R^2 is again near 0.06.

Predictive utility does not clear the floor. The learned free read-out reaches $R^2 = 0.06$ (95% CI $[-0.03, 0.13]$), so its lower bound falls below the registered minimum of zero. The paired comparison against the untrained encoder is more telling: training changes held-out R^2 by -0.02 (95% CI $[-0.04, 0.00]$), slightly negative and consistent with no gain (Figure 2a). The self-consistency oracle recovers the target almost perfectly ($R^2 \approx 1.00$), confirming that the target is reconstructible and that the failure lies with the learned features, not the metric. Measured nuisances explain essentially nothing alone, and adding them to the learned features leaves held-out R^2 unchanged.

Reflection augmentation does not help. On the primary lane the reflection-augmented encoder differs from vanilla by 0.00 in source-balanced R^2 (95% CI $[-0.01, 0.01]$). The augmentation that most directly encodes the symmetry has no measurable effect on the recovered signal.

The construction works, but the encoder is not why. The constructed odd/zero-origin read-out is exactly antisymmetric for every prediction and seed, so its oddness gate passes at the 10^{-6} tolerance. Its absolute predictive value is weak ($R^2 = 0.04$, 95% CI $[-0.04, 0.11]$), and the same construction on an untrained encoder does at least as well: learned minus untrained is -0.06 (95% CI $[-0.10, -0.02]$) under the odd/zero construction and -0.07 (95% CI $[-0.11, -0.03]$) under odd/free, both excluding zero on the negative side. Whatever modest structure the odd lanes use belongs to the parity wrapper on the anatomical pairs, not to the learned representation. Training nudges only the even part ($+0.02$, 95% CI $[0.01, 0.04]$)—the opposite of where a laterality signal lives.

The native read-out is far from antisymmetric. Unconstrained, the learned free read-out is not sign-reversing: its normalized native output antisymmetry error is 0.22 (95% CI $[0.19, 0.24]$), well above the 0.10 margin and barely different from the untrained encoder’s. Exact antisymmetry appears only when imposed.

Training degrades reflection equivariance. The strict token test is the sharpest result. The learned representation is not equivariant, with $q = 0.11$ (95% CI $[0.10, 0.14]$) above the 0.10 margin, while the untrained initialization is closer at $q = 0.08$ (95% CI $[0.07, 0.09]$). The learned-minus-initial difference is $+0.03$ (95% CI $[0.02, 0.05]$), so training moves the tokens away from the approximate equivariance they started with (Figure 2b). Reflection augmentation shrinks the damage

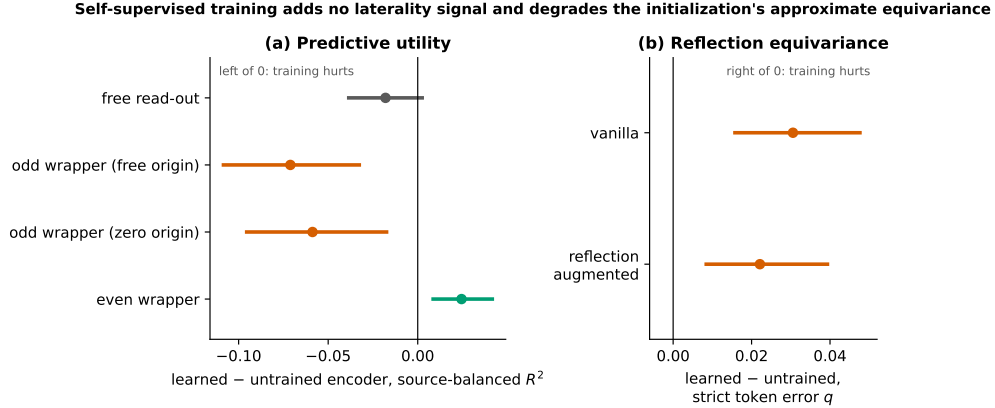


Figure 2: What self-supervised training changed, as learned minus untrained differences with 95% source-cluster bootstrap intervals. (a) Predictive utility: for the free read-out and both odd (antisymmetric) constructions the difference is at or below zero, so training adds no usable laterality signal. (b) Reflection equivariance: the strict token error rises after training for both variants, so training moves the representation away from the equivariance present at initialization.

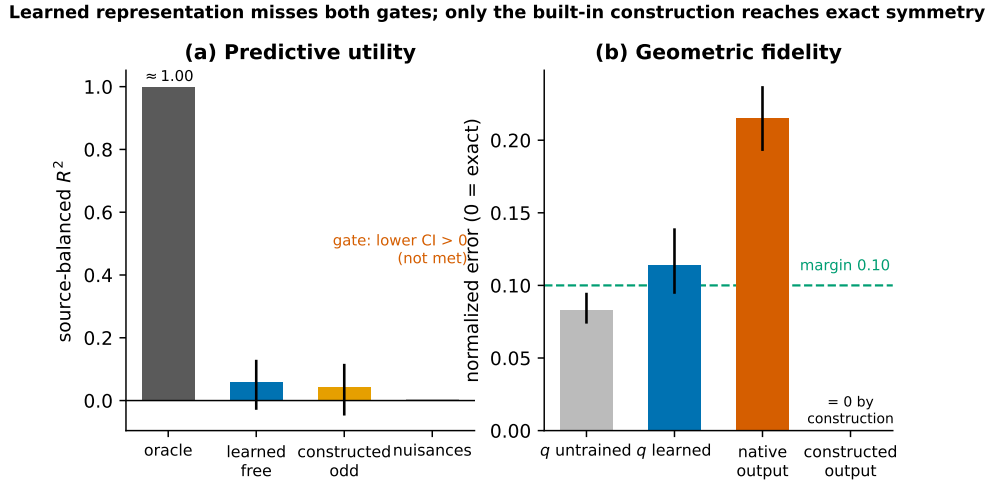


Figure 3: Absolute levels against the frozen thresholds. (a) Source-balanced R^2 : the self-consistency oracle is near one, while the learned free read-out and the constructed odd read-out have intervals crossing zero. (b) Geometric fidelity: the learned token error and the native output antisymmetry error both exceed the 0.10 margin, and the untrained encoder is closer to equivariant than the learned one; only the imposed construction reaches zero.

194 a little ($q = 0.11$, learned minus initial $+0.02$, 95% CI $[0.01, 0.04]$) but neither reaches the margin
 195 nor beats the initialization. Figure 3 places the learned representation against the frozen thresholds
 196 on both axes: it misses the predictive-utility gate and both fidelity margins, and only the built-in
 197 construction reaches exact symmetry. Table 2 collects the decisions.

198 10 Uncertainty and decision rules

199 Optimization-seed variation and source-sampling uncertainty stay separate. Five seeds show how
 200 results shift across registered initializations; 2,000 bootstrap repetitions resample source videos—
 201 keeping every sequence and seed from a source together—to show how the estimate shifts over
 202 the observed collection. These intervals are conditional on the fitted cross-validation pipeline
 203 and do not propagate new-seed, split-allocation, or full-retraining uncertainty, nor do they turn

Table 2: Registered decision gates and outcomes. Every empirical gate is unmet; the only gate that passes is a property of the construction.

Registered gate	Requirement	Outcome
Native probe predictive utility	learned R^2 lower CI > 0	not met (0.06, $[-0.03, 0.13]$)
Native training content	learned – untrained lower CI > 0	not met (-0.02 , $[-0.04, 0.00]$)
Native output symmetry	error upper CI < 0.10	not met (0.22, $[0.19, 0.24]$)
Strict token equivariance	learned q upper CI < 0.10	not met (0.11, $[0.10, 0.14]$)
Equivariance training effect	learned – initial q upper CI < 0	not met ($+0.03$, $[0.02, 0.05]$)
Learned increment over nuisances	lower CI > 0	not met (0.05, $[-0.02, 0.12]$)
Constructed predictive utility	learned R^2 lower CI > 0	not met (0.04, $[-0.04, 0.11]$)
Constructed training content	learned – untrained lower CI > 0	not met (-0.06 , $[-0.10, -0.02]$)
Constructed output oddness	error $\leq 10^{-6}$, every seed	met (by construction)

the 93 videos into a random sample of people or recording conditions. Every threshold in `config/protocol.json` was frozen before results were seen, and the protocol fixes a null-result policy: a failed diagnostic is reported as is and never licenses retuning. That policy is what makes Section 9 usable rather than a starting point for search.

11 Limitations

The split prevents source-video overlap but cannot guarantee distinct people across videos, so the supported claim is post-development, within-GAVD cross-validated performance on held-out source videos. It is not a claim about unseen people, another dataset, clinical validity, prevalence, diagnosis, or treatment. The target captures one signed left-versus-right motion contrast, not the full kinematics of gait, and is not a validated clinical index. The strict test fixes a particular joint permutation and identity channel action, so a different notion of equivariance could give a different number. All findings remain conditional on the BlazePose schema, the pose-extraction system, the preprocessing, the architecture, the registered seeds, and the measured controls; a larger model, a different objective, or a denser target might behave differently, and the null is specific to the configuration we registered. Unmeasured acquisition and missingness effects may remain.

12 Ethics and data use

Public availability of video links is not authorization for research use or redistribution. The suite stores no raw video and no identity-bearing frame, but derived poses, source identifiers, embeddings, predictions, and checkpoints can still be linkable or sensitive. The institutional ethics determination, the data-use review, and the derived-pose release review are unresolved, and submission and artifact release stay blocked until each carries a genuine dated internal reference. A careful statistical design does not substitute for those determinations.

13 Reproducibility

Notebook 00 freezes the protocol and reports governance; 01 audits cohort construction and the target; 02 builds the source-level folds; 03 trains the fold-local encoders; 04 fits read-outs and produces held-out predictions; 05 aggregates checkpoints and uncertainty; 06 gates a possible external dataset that the GAVD review cannot authorize on its own. Every number in Section 9 traces to report artifacts under `artifacts/paper/protocol_<digest>/report/` carrying matching protocol, context, cohort, split, checkpoint, and implementation digests, and the figures are regenerated from those artifacts by `docs/figures/make_v21_figures.py`. Split assignments live in `source_splits.json`, the rules in `config/protocol.json`, and the interpretation boundary in `PROTOCOL.md` and `RUNBOOK.md`.

References

GAVD project. Gait abnormality video dataset: Data access and responsible-use notes. GitHub repository, 2026. URL <https://github.com/Rahmyyy/GAVD>. Accessed: 2026-09-03.

239 Rahm Ranjan, David Ahmedt-Aristizabal, Mohammad Ali Armin, and Juno Kim. Computer vision for
240 clinical gait analysis: A gait abnormality video dataset. *IEEE Access*, 13:45321–45339, 2025. doi:
241 10.1109/ACCESS.2025.3545787.